

YICHEN (ANNA) TU

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EDUCATION

Duke University, Pratt School of Engineering, Durham, USA

Expected May 2027

Master of Engineering, Financial Technology

- **Relevant Coursework:** Programming (C++, Python), Software Engineering (C++, Python), Machine Learning, Quantitative Risk Management

University of St Andrews, St Andrews, Scotland

September 2021 – May 2025

Bachelor of Science (Honours) in Economics and Mathematics

- **Relevant Coursework:** Causal Inference, Time Series Analysis, Mathematical Modeling, Advanced Linear Algebra

University of Virginia (Non-degree, Exchange Student), Charlottesville, USA

August 2023 – May 2024

- **Relevant Coursework:** Real and Complex Analysis, Econometrics, Stochastic Processes, Ordinary Differential Equations/Partial DE
- **Technical:** C++, Python (pandas, numpy, scikit-learn), R Studio, SQL, MATLAB, Latex

PROFESSIONAL EXPERIENCE

Fortune SG Fund Management

Shanghai, China

Global Markets Intern (Investment Department)

June – August 2025

- Built an Excel pipeline to analyze ETF volatility and diversification risk, visualized factor contributions for risk reporting.
- Synthesized analysis for fund presentations on HuaAn Oil & Gas LOF (162411.SZ) by tracking NAV/price performance and linking movements to OPEC+ output and market news.
- Extracted key risk drivers from 10+ sell-side roadshows on HK/US equities and macro, informing portfolio assessments.

Hwabao Securities

Shanghai, China

Mutual Fund Research Intern (Research and Innovation Department)

June – August 2024

- Analyzed ~50,000 raw data points using multivariate regression and identified operational risk factors that significantly influenced process time, including the impact of returned data and differences across various stages of approval.
- Utilized Python to process index stock weights and factor exposure datasets for 9 indices, calculated factor exposures for ~1,000 data points per stock, and leveraged Excel to visualize key insights for leadership decision-making.
- Built Python data pipeline to prepare multi-quarter financial data for quantitative risk modeling and strategy evaluation.

PROJECTS AND RESEARCH EXPERIENCE

MEV Bot Detection on Ethereum Mainnet

Jan – May 2026

- Derived 19 behavioral features from 1.50M Ethereum transactions across 481,964 addresses; validated feature discriminability via Mann-Whitney U tests and Cliff's delta.
- Trained LR, RF, and XGBoost under 121:1 class imbalance with label augmentation; final XGBoost achieved OOF AUC = 0.9833, Precision@100 = 0.99, and identified 6/10 top sandwich-attack extractors on external holdout.

Volatility Forecasting on QQQ

Jan – May 2026

- Modeled conditional volatility using EWMA, GARCH(1,1), and GJR-GARCH(1,1) on 26 years of QQQ daily returns; GJR-GARCH reduced out-of-sample MSE by ~8% vs. EWMA over a 1,323-observation expanding window.
- Backtested VaR and Expected Shortfall at 95%/99% confidence levels; GJR-GARCH achieved lowest exceedance rate (5.74%) and highest Sharpe-like ratio (0.052) under a volatility-targeting strategy.

Honours Dissertation – Standardization and IPW for Causal Inference

September 2024 – May 2025

- Translated a public health research question into a rigorous statistical framework: estimated causal effect of smoking on overweight incidence using BRFSS 2023 (~400,000 obs) via standardization and IPW; ATE ranged from -5.5% to -9.3% across parametric and non-parametric methods.

Research Assistant - Business School

September – December 2024

- Built a scalable Python data pipeline to link LinkedIn employment records, patent inventor, and assignee datasets via fuzzy matching and probabilistic record linkage
- Improved entity-resolution accuracy by 5–17% across 1TB+ of data, enabling faculty research on corporate innovation.

STEP Diving into Data 2024

May – July 2024

- Engineered features from ADHD dataset (3,000+ rows) via imputation, categorical encoding, and numeric parsing; built multifactor regression model to quantify environmental risk factors (metal exposure, soil concentration) on ADHD incidence using R.
- Led a team of 7 across full project lifecycle, from data cleaning and feature engineering to model interpretation and mentor presentations.